

MARCUS NIIHAU-ASHBY

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EDUCATION

University of California, Berkeley

Expected May 2026

Bachelor of Arts in Computer Science, Bachelor of Arts in Cognitive Science • GPA: 3.5/4.0

Berkeley, CA

- **Relevant Coursework:** Data Structures and Algorithms/Programming Methodology, Efficient Algorithms and Intractable Problems, Principles and Techniques of Data Science, Foundations of Data Science, Structure and Interpretation of Computer Programs, Computational Models of Cognition, Linear Algebra, Calculus, Discrete Mathematics and Probability Theory, Introduction to Internet Architecture/Protocols, Neuroethology: Complex Animal Behavior and Brains, Linguistic Science
- **Activities:** Head Lecturer of Hawaiian Language Course, Full Stack @ Berkeley, Motorcyclists of Berkeley, Surf Club, Climbing

WORK EXPERIENCE

Project Manager / Software Engineer

August 2025 - Present

Neurotech at Berkeley

Berkeley, CA

- Identified patterns of neural coherence in meditating monks by modeling 64-channel EEG data with temporal analysis
- Developed self-simplifying LLM loops for recursive model refinement, inspired by neural dynamics from EEG meditation data
- Decoded Pink Floyd's "Another Brick in the Wall" from auditory cortex signals via the Knight Lab's iEEG pipeline

Software Engineer Intern

January 2025 – August 2025

Gameheads Oakland

Oakland, CA

- Lead a 6-person team as the sole engineer, owning the complete technical lifecycle from architecture to deployment
- Developed custom Unity tools, including an audio manager and dialogue system, to accelerate a non-technical team's workflow
- Implemented critical playtest feedback by writing custom shaders, redesigning the UI, and overhauling level layouts
- Presented weekly technical progress to industry directors and engineers from Xbox, Double Fine, and Amazon Music
- Placed in the top 10% of 25 teams with 2 award nominations for 'Best Programmer' and 'Best Narrative Designer'

PROJECTS

Kapu Aloha — Published Video Game | C#, Blender, Unity Engine

Jan. 2025 - Aug. 2025

- Rapidly mastered C# and the Unity engine from scratch as the sole programmer, writing over 8,000 lines of production code
- Designed and implemented a seamless teleportation system to create an optical illusion, managing player and camera transforms
- Engineered the core gameplay loop, including a multi-state enemy AI and an environment-altering dynamic inventory system
- Optimized rendering performance by opting for orthogonal 2D sprite-based assets, and subsequently modeling them in Blender

Meta Breaker — Data Analysis for League of Legends | Python, Pandas, Riot API, React, JavaScript, PostgreSQL

Mar 2025 - Present

- Discovered winning, "off-meta" strategies in League of Legends by analyzing over 1M matches with Python and PostgreSQL
- Engineered efficient data pipelines to transform raw game data into structured insights, discovering 1,000+ strategies
- Now trusted by creators with a combined audience of 500k+ players after demonstrating significant time savings
- Engineering a React/Node.js dashboard to provide creators/users with visual insights into strategy effectiveness

Movie Classifier — Machine Learning Algorithm to parse screenplays to genres | Python, SQL, NumPy

Oct. 2024 - Dec. 2024

- Trained a K-Nearest Neighbors model that predicts a movie's genre just by reading its script (trained on 5,000+ films)
- Optimized model accuracy through NLP preprocessing techniques like stemming and word-frequency vectorization
- Deployed train/test validation framework and feature standardization with NumPy, further improving model accuracy

Markizzy.com — Personal Landing Page | HTML, Tailwind CSS, JavaScript, Next.js, Node.js

Mar. 2025

- Designed and deployed a professional portfolio site, leading to 50 monthly website visits since deployment
- Mastered TailwindCSS V4 to build a responsive site with a Node.js backend, achieving a 99% PageSpeed score
- Implemented modular architecture, resulting in 50% faster update cycles and simplified maintenance workflows

Gitlet — Git created from scratch | Java

June 2024

- Re-engineered Git's core commit and storage system from scratch using Java serialization and SHA-1 hashing.
- Enabled features like branching/merging to execute in $O(\log N)$ by engineering a 3-tree architecture to track commits
- Built a personalized CLI, including robust edge-case handling procedures found through test-driven development

SKILLS

Languages: Python, Java, SQL, C, C#, Lisp, MATLAB, Swift, HTML, CSS, JavaScript, TypeScript

Tools/Libraries: pandas, NumPy, Matplotlib, PyTorch, PostgreSQL, MySQL, AWS, BeautifulSoup, Blender, Maya, Unity Engine

Frameworks: React, React Native, Next.js, Node.js, Tailwind CSS, Django